

Use Case Name:	Manage Library Catalog	
Scenario:	Librarian maintains the catalog database, while Students and Faculty search, view, and select library resources.	
Triggering Event:	Student/Faculty accesses the catalog search interface to find materials, or a Librarian accesses the dashboard to manage inventory.	
Brief Description:	Encompasses catalog searching, filtering, and viewing item availability by students/faculty (View Library Catalog), as well as administrative CRUD operations (add, edit, delete) performed by the Librarian. Interfaces with Koha ILS and DocuPulse.	
Actors:	Student/Faculty, Librarian, Koha ILS (External System), DocuPulse (External System)	
Related Use Cases:	Includes: View Library Catalog. Extends: Manage Borrowed Books.	
Stakeholders:	• Student/Faculty: To easily find and view required resources. • Librarian: To maintain accurate library records. • DocuPulse: To synchronize external metadata.	
Preconditions:	• Student/Faculty successfully signs in to LIRA, OR Librarian successfully signs in to Koha. • Catalog database and LIRA portal are online.	
Postconditions:	• Matching catalog items and availability details are displayed. • Catalog database is updated, saved, and synchronized with Koha ILS and DocuPulse.	
Flow of Activities:	Actor	System
	1. Student/Faculty successfully signs in to LIRA and accesses the catalog search interface.	1.1 System validates session and loads search interface.
	2. Student/Faculty enters search criteria (e.g., title, author, keyword) and submits.	2.1 Queries local database records and Koha ILS to retrieve matching items. 2.2 Displays a list of matching catalog items.
	3. Student/Faculty selects a specific item from search results.	3.1 Displays detailed item information, availability, and location.
	4. Librarian successfully signs in to Koha and accesses the catalog management dashboard.	4.1 Displays the current catalog list and data entry forms.
	5. Librarian submits new item details, edits existing records, or confirms item deletion.	5.1 Validates submitted data, updates catalog database, and synchronizes with Koha ILS. 5.2 Displays success confirmation message.
Exception Conditions:	2.1 If no items match search criteria, the system displays a 'No results found' message. 5.1 If data validation fails during an update (e.g., missing ISBN), the system highlights errors and prompts for correction. 5.2 If an item being modified or deleted is checked out, the system invokes Manage Borrowed Books to resolve status conflicts.	

Use Case Name:	Manage Borrowed Books	
Scenario:	Students and Faculty initiate borrow requests, while the Librarian manages inventory lifecycle circulation and status tracking.	
Triggering Event:	User initiates a borrow request while viewing an item, or a Librarian processes check-ins, renewals, or returns.	
Brief Description:	Handles user book borrowing requests (Borrow Books) and administrative circulation management (Manage Borrowed Books). Communicates with Koha ILS to manage loan statuses, due dates, and inventory lifecycle.	
Actors:	Student/Faculty, Librarian, Koha ILS (External System)	
Related Use Cases:	Extension of: Manage Library Catalog. Includes: Borrow Books.	
Stakeholders:	• Student/Faculty: To successfully acquire and borrow books. • Librarian & Koha ILS: To track inventory lifecycle and circulation.	
Preconditions:	• Student/Faculty successfully signs in to LIRA with a valid account, OR Librarian successfully signs in to Koha. • Selected item is available and LIRA is connected to Koha API.	
Postconditions:	• Item status is updated to 'Checked Out' or returned, user account reflects loan, and transaction is logged in Koha ILS.	
Flow of Activities:	Actor	System
	1. Student/Faculty successfully signs in to LIRA and initiates a borrow request for an available item.	1.1 Verifies user account status and borrowing limits. 1.2 Prompts user for final confirmation.
	2. Student/Faculty confirms borrow request and accepts terms.	2.1 Transmits transaction data to Koha ILS to log checkout. 2.2 Updates local item status to 'Checked Out' and displays borrowing confirmation receipt and due date.
	3. Librarian successfully signs in to Koha and accesses circulation module.	3.1 Queries Koha and displays synchronized status of borrowed books. 3.2 Processes check-ins, renewals, or lost records and commits updates.
Exception Conditions:	1.1 If the user has a blocked account or overdue items, the system denies the request and displays a warning. 2.1 If the item becomes unavailable during the transaction, the system cancels the process and notifies the user. 2.2 If Koha ILS is unreachable, the system queues the transaction locally and alerts the Librarian.	

Use Case Name:	Manage AI Chatbot
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Scenario:	Students and Faculty ask AI questions for immediate research assistance, while the Librarian with admin privileges configures the AI knowledge base and parameters.	
Triggering Event:	User submits a query to the chat widget, or a Librarian accesses the AI configuration panel.	
Brief Description:	Encompasses automated response generation for student research queries (Ask AI Chatbot) and administrative tuning of chatbot parameters, custom prompts, and knowledge sources by the Librarian (Manage Chatbot).	
Actors:	Student/Faculty, Librarian, AI Engine	
Related Use Cases:	Includes: Ask AI Chatbot.	
Stakeholders:	• Student/Faculty: For immediate, accurate research assistance. • Librarian: To audit AI interactions and ensure response safety and accuracy.	
Preconditions:	• Student/Faculty successfully signs in to LIRA, OR Librarian successfully signs in with admin privileges. • AI engine and RAG pipeline are online.	
Postconditions:	• Relevant research response is generated and displayed, OR chatbot configuration and knowledge base are successfully updated.	
Flow of Activities:	Actor	System
	1. Student/Faculty successfully signs in to LIRA and submits a question into the chat widget.	1.1 Parses query and routes it to AI Engine.
	2. AI Engine generates synthesized response based on RAG prompt context.	2.1 Formats retrieved answer, displays it in chat interface, and logs interaction.
	3. Librarian successfully signs in with admin privileges and accesses chatbot configuration panel.	3.1 Displays current knowledge sources, prompts, and interaction logs.
	4. Librarian uploads new knowledge documents or edits custom prompts.	4.1 Validates and applies new configurations to AI engine.
Exception Conditions:	1.1 If AI service times out, the system apologizes and provides contact information for a human librarian. 2.1 If AI generates a response violating safety policies, the system intercepts it and substitutes a safe default message. 4.1 If Librarian uploads an incompatible document, the system rejects it and displays an error message.	

Use Case Name:	Manage Reports
Scenario:	The Librarian generates analytics regarding library usage, circulation data, and chatbot interactions.

Triggering Event:	Librarian requests a statistical or operational report from the dashboard.	
Brief Description:	Allows the Librarian to compile, preview, adjust, and export data reports by extracting data from local LIRA databases and integrating circulation data from the Koha ILS.	
Actors:	Librarian, Koha ILS (External System)	
Related Use Cases:	N/A	
Stakeholders:	Library Management: To make data-driven decisions regarding acquisitions and system usage.	
Preconditions:	- Librarian successfully signs in to Koha and holds administrative reporting privileges. - Log data exists in both LIRA and Koha databases.	
Postconditions:	Formatted report is generated, previewed on dashboard, and available for download (PDF/Excel).	
Flow of Activities:	Actor	System
	1. Librarian successfully signs in to Koha and selects report type and date range from dashboard.	1.1 Requests necessary usage data from local database and circulation data from Koha ILS.
	2. Koha ILS provides requested raw circulation data.	2.1 Aggregates combined data and displays report preview on dashboard.
	3. Librarian adjusts parameters or clicks to export final report.	3.1 Generates downloadable file (PDF/Excel) and provides it to user.
Exception Conditions:	1.1 If Koha ILS is offline, the system generates a partial report using only local LIRA data and appends a warning note. 2.1 If the selected date range contains no data, the system outputs an empty report accompanied by a notification.	